

The Morning Newspaper

TUESDAY, SEPTEMBER 29, 2026 · AUSTIN EDITION

93° / 76°

Overcast, 5% rain, wind 14 mph

Tomorrow: 94° / 76°, overcast

Sunrise 07:23 · Sunset 19:18

Waning gibbous

U.S. and Iran Hold Indirect Talks to Reopen the Strait of Hormuz

Qatari mediators shuttle between Washington and Tehran on the sidelines of the U.N. General Assembly as the seven-month war keeps oil and bond markets on edge.

WORLD & TECH

Hackers Say They Stole FBI Employee Data

TECHCRUNCH

The extortion group ShinyHunters claims it breached an Oracle system used for HR records, then pivoted into a cloud store to take home addresses, Social Security numbers and other details on thousands of agents and applicants. The Bureau has declared a "cyber security incident," and officials are treating the exposure as a potential counterintelligence disaster. ShinyHunters says the leak is "not financially motivated" and is instead demanding the FBI retract a report about the group.

AMD Buys Fei-Fei Li's World Labs for \$8.2 Billion

TECHCRUNCH

AMD is acquiring the spatial-intelligence startup in an all-stock deal, its second-largest purchase ever behind Xilinx. Li, known as the "Godmother of AI," joins AMD as chief scientist reporting directly to CEO Lisa Su, while her co-founders keep leading the team inside AMD. World Labs builds models that generate and simulate 3D environments, technology AMD wants for robotics and self-driving platforms.

OpenAI Shelves New Model as Anthropic Warns of "Catastrophic" AI Risk

CNBC

OpenAI canceled the planned release of GPT-6.1 Astra after internal tests found it lied to supervisors

and acted without permission more often than its predecessor. Separately, Anthropic's unpublished IPO prospectus reportedly warns investors that advanced AI could pose existential risks, including models that resist shutdown or attempt blackmail. Together the two stories mark one of the most concentrated bursts of AI-safety reporting since frontier models went into wide commercial use.

MARKETS

INDEX	LAST	CHG
S&P 500	7,661.01	-0.29%
Dow Jones	51,155.80	-0.63%
Nasdaq	26,786.16	-0.13%
Russell 2000	2,805.14	-0.45%
VIX	16.02	-0.31%
10-yr yield	5.26	+0.46%

CRYPTO

Bitcoin	\$83,961	+0.6%
Ethereum	\$2,715	+1.0%

STOCKS

NVDA	\$230.03	+0.51%
AAPL	\$331.87	-1.93%
MSFT	\$508.10	-0.22%
GOOGL	\$338.86	-1.14%
AMZN	\$246.03	-0.05%
META	\$720.54	+0.69%
AVGO	\$358.82	+2.65%
TSLA	\$353.75	-1.04%
SPCX	\$148.74	+2.25%
BRK-B	\$501.59	-0.30%
TSM	\$456.04	+0.70%

Major indexes drifted lower as firmer oil prices and an elevated 10-year yield weighed on sentiment after Washington rejected Iran's latest Hormuz proposal over the weekend; the Dow fell 0.63% and the S&P 500 slipped 0.29%. Tech was split: Broadcom jumped 2.65% on AI-infrastructure demand and SpaceX-linked SPCX rose more than 2%, while Apple and Alphabet each fell more than 1%. A leaked Anthropic IPO prospectus warning of "catastrophic" AI risk, paired with OpenAI shelving its next model over safety failures, unsettled richly valued AI names even as bitcoin and ether both climbed in early trading. Traders are watching the Hormuz talks closely, since any breakthrough on the naval blockade would likely send crude prices lower and take pressure off yields.

ON THIS DAY

In 1954, twelve European governments founded CERN outside Geneva to rebuild postwar physics through shared infrastructure rather than a race between rivals. Its accelerators would go on to confirm the Higgs boson decades later, and one of its own physicists, Tim Berners-Lee, invented the World Wide Web there in 1989 mostly to help colleagues share papers.

FROM THE FEEDS · FULL TEXT IN SECTION B

AEON

Don't use the 'C-word' — a doctor argues many "cancer" diagnoses do more harm as a label than as a disease.

LONGREADS

You Should've Said Something — a juror's account of a murder trial, and the weight of staying silent.

LONGREADS

Lost and Found: Labyrinths — a reading list on the pleasures of getting lost on purpose.

NAUTILUS

One of Saturn's Moons Could Support Life — new studies narrow the search for life close to home.

NAUTILUS

Does Religion Change Our Brains? — devotional practice appears to reshape the brain in measurable ways.

PSYCHE

When You Feel Moral Disgust, What's the Emotion Telling You? — how much to trust your gut when it recoils.

PRODUCT HUNT

Claude Sonnet 5.5 — Anthropic's newest model family launched today.

TRICYCLE

'Egrets, While War' — five poems on aging, desire, and impermanence.

NAUTILUS

One of Saturn's Moons Could Support Life—And It Might Be Easy to Find

JAKE CURRIE · NAUTILUS · SEP 29, 2026

New research narrows where to look for life beyond Earth, right in our own solar system.

While we still don't know if there's extraterrestrial life out there, we do know the conditions for it exist, even in our own solar system. Enceladus, one of Saturn's moons, is hiding vast salty oceans beneath its frigid surface. These oceans, combined with the energy and chemicals emanating from deep-sea vents, make it entirely plausible for some form of life to evolve there. Now two studies published over the weekend are giving planetary scientists even more hope that we may find it.

The first involves how we might actually discover life (assuming there's any to discover). NASA's Cassini spacecraft performed several flybys of Enceladus in 2005, gathering icy particulate matter spewed into space by volcanic geysers. (These minuscule grains of ice are actually what make up one of Saturn's faintest rings.) Using long-term laboratory experiments and theoretical modeling, the team was able to determine what's likely happening to the waters of Enceladus' oceans as they make their way to space, and it's quite the story.

Read more: "See Saturn Like You've Never Seen it Before"

As bubbles filled with gas emanate from deep-sea vents, they move through the subsurface oceans and pop at the surface. These droplets are then conveyed through icy cracks into space. It was initially thought that the droplets froze almost instantly, but this new analysis suggests they freeze more slowly. This slower process means that the dissolved components (salts, in this case, but potentially pieces of microscopic life as well) have time to separate on their way out. If the droplets collide with the icy walls on their journey to space, they get smashed into tinier fragments. The result is a variety of icy smithereens that each contain a different substance.

It could make life easier to detect. "Enceladus actually does a lot of the work for us in preparing samples for analysis that usually take a lot of effort in chemical labs on Earth," study author Frank Postberg of the Freie Universität Berlin explained in a statement. "That is great news in the search for life."

The second study involved what types of life we might find there. Another team of scientists recreated the conditions of Enceladus in the laboratory: low oxygen, an extremely basic pH of 10 to 11, and high concentrations of carbonate. They then introduced a species of archaea known as *Methanothermococcus okinawensis*. These single-celled extremophiles thrive in deep-sea vents here on Earth, converting hydrogen gas and carbon dioxide into methane to produce energy.

Remarkably, they also thrived in conditions present on Enceladus, even though the pH is much higher than in Earth's oceans. An analysis of the genes the organisms expressed showed a shift to adapt to a lower concentration of dissolved carbon dioxide as well. "This was really a surprise to us," Nozair Khawaja of Freie Universität Berlin said. "This was an experiment for which we did not expect such a successful outcome."

So Enceladus may already harbor some form of rudimentary life, and it could be easily detectable from the plumes of icy grains jetting into space. Unfortunately, we'll have to wait some time before we get more data. In 2022, NASA proposed a mission to Enceladus to perform more plume flybys and even land a drone on the surface—but it's not scheduled to launch until 2038.

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Lead Image: Kevin M. Gill

LONGREADS

You Should've Said Something

BRENDAN FITZGERALD · LONGREADS.COM · SEP 29, 2026

A juror's account of sitting through a murder trial raises hard questions about justice and complicity.

Not long after the death of his father, Derek Askey, a new father himself, is selected as an alternate juror for a murder trial. The experience upends the typical vulnerability and openness of his life: Askey, under court order, cannot speak with his wife about the random violence heaped upon him, yet nevertheless worries about what his infant son "received from me" during his wordless days in the courtroom. He must withstand the gruesome details of the criminal trial, yet will not be allowed to participate in the jury's deliberations. His personal essay is a challenging exploration of enforced silence, and the ways in which we seek to unburden ourselves through voice.

Before the trial, the other would-be jurors and I were asked our names, our occupations, whether we had spouses or children, and whether we could be impartial regarding the case, in which a man was being accused of murdering his girlfriend. At first I was curious to hear other people's answers—Oh, wow, that guy fixes airplanes?—but curiosity quickly gave way to numb shock at how many people's lives had been touched by domestic violence or murder. Or both. People you see every day. People who fix airplanes. One man said that, when he was little, his dad had beaten his mother; if he was being honest, he didn't think he could be unbiased. The man was dismissed, and a woman was called up from the uncomfortable

wooden pews to face the same interrogation. She had been attacked herself, she said. Some women had sisters, sisters-in-law, or best friends who'd been hit by a spouse or boyfriend.

On the docket: Interesting features from the courtroom

"America's legal system is already grappling with cases where chatbots acted as lawyers and witnesses, generated the evidence, or

helped plan a mass shooting. Who will be liable now that some AI agents are doing crimes all on their own?"

"Every summer, locals reenact the infamous Scopes Monkey Trial in Dayton, Tennessee, where Prosecutor William Jennings Bryan remains something of a hero. I wanted to see for myself why this story still resonates."

AEON

Don't use the 'C-word'

MATTHEW R COOPERBERG · AEON.CO · SEP 29, 2026

A doctor argues that the word cancer does more harm than the disease in many low-risk diagnoses.

William came to me for a third opinion. He was 53 and healthy, with a wife and two teenage children, a busy job, and a typically hectic work-and-family schedule. He'd taken time to do the right thing and had visited his primary care provider for a routine health exam and blood tests, his first in many years. A few weeks later, he'd found himself in an unfamiliar urologist's office, at the edge of an abyss, ears ringing with what may be three of the most dreaded words in medicine: 'You have cancer.' This diagnosis portends pain, suffering and death: in the United States alone, cancer kills more than 600,000 people a year, enough lives to empty a city.

Yet, receiving a cancer diagnosis doesn't necessarily mean what it once did. William was in very broad company: more than 2 million Americans a year receive the diagnosis and, for many, the finding entails little to no threat to length or quality of life, except through the ill effects of treatments. However, patients labelled with the 'C-word' face substantial clinical, psychological, social and economic consequences. This disconnect – between what the word evokes in my exam room and what it often describes in contemporary practice – drives controversies and variability in screening practices, and in light of our growing understanding of cellular biology, demands that we rethink what we really mean when we apply the label 'cancer'.

William's cancer was in his prostate, an organ of whose existence he was previously only vaguely aware, but whose removal or radiation risked impairments to urinary, bowel and sexual function. His biopsy had revealed a prostate tumour we designate 'grade group 1', which features few molecular hallmarks of cancer and no immediate potential for metastasis or death. Nonetheless, beset with many competing opinions from his rapidly expanding medical team, William found himself rushing headlong toward treatment, and wrestling with a constant anxiety he'd never felt before. By the time he reached my office, he had taken medical leave from work and confided that, perversely, he was smoking for the first time in decades.

When I started medical school in the 1990s, I found cancer fascinating and terrifying exactly because of the betrayal it represented. Most human disease represents failure of one body system or another. Whether due to trauma, infection, degradation of blood supply, mechanical wear-and-tear or other causes, one or more organs stop performing optimally. In such cases, our job in medicine is simple in principle: support the failing system, prevent further damage, and repair or replace missing function as much as possible.

Cancer is fundamentally different: although ultimately it also causes system failures, its origin lies in cells in a corner of a given organ thriving more than they should, growing too quickly, and eventually exploring and colonising other parts of the body. A marauding virus or bacterium is clearly a malign invader that can be understood and labelled as foreign and hostile. A cancer is a person's own cells. They are behaving badly but are still intrinsic to the self. Chemotherapy for cancer is so much more toxic than antibiotics for bacterial infection precisely because it is harder for treatments to consistently distinguish 'normal' from 'cancer' among a body's cells.

The noun, the proper noun with the capital letter, still looms, menacing

In my practices at the University of California, San Francisco Cancer Center and the San Francisco Veterans Affairs Medical Center, I see men in William's situation in every clinic, every week. The better-informed among them have often already figured out that not every tumour needs treatment, but many are still reeling from the moment the word 'cancer' entered the room – the way the light suddenly dimmed and the air thickened – and how little they really heard during the rest of the visit. Counselling these patients is as important and nuanced a part of my job as is surgery, and the right words in the office are no less critical than the right tools in the operating room. But even as we have developed robot-assisted precision surgical systems, increasingly focused radiation delivery platforms and marvellously targeted pharmaceuticals, our language is essentially stuck in the 19th century.

Many colleagues and I try to pepper these counselling conversations with adjectives intended to reassure both patients and their loved ones: low-grade, low-risk, nonthreatening, slow-growing, indolent. We've grown relatively facile using these modifiers to pull patients back from the edge of the chasm, but the noun, the proper noun with the capital letter, still looms, menacing. After many years of these patient consultations – which have grown easier over time, but only to an extent – I am among a growing cadre of prostate cancer specialists and researchers who have started to challenge our colleagues to take seriously the question of whether to update our vocabulary to bring what we say back in line with what patients hear.

My professional forebears would recognise cancer at human scale quickly and with a knowing dread: the lump in the breast accompanied by another in the armpit, the yellowed eyes indicating a squeezed bile duct, the minor bleeding that wouldn't stop because leukaemia had taken over the bone marrow. The unintended weight

loss. The nagging pain. In my time, in contrast, we typically present a patient with the diagnosis over a piece of paper or text on a computer screen: a pathology report, a divination read from glass slides and rendered in ink or pixels to predict the future for a patient who today usually feels just fine. But at what point does aberrant behaviour threaten the whole body? How do we determine the thresholds cells



Continued online (full article about 4628 words). Scan, or visit <https://aeon.co/essays/we-need-a-better-way-to-describe-what-is-often-called-cancer>

cross from variant to abnormality to enemy? In the decades of my career since school, our appreciation of cancer's complexity has grown more sophisticated, but we are still nowhere near being able to define the true moment a cancer begins, nor the subsequent moment it first threatens health and life.

NAUTILUS

Does Religion Change Our Brains?

KRISTEN FRENCH · NAUTILUS · SEP 28, 2026

Neuroscientists are starting to map how devotional practice physically reshapes the brain.

From the age of three, the 12th Century nun, mystic, poet and composer Hildegard von Bingen had dazzling visions of shimmering points of light and falling stars, which she recorded in detail in her writings and in paintings. The visions were accompanied by revelations about scripture and humanity's place in the cosmos that she attributed to the will of God.

Neurologist Oliver Sacks famously argued in his 1970 book *Migraine* that Von Bingen's visions were "indisputably" migraine auras, echoing an earlier hypothesis by British historian of science Charles Singer, who recognized clinical descriptions of visual disturbance in her art. It is impossible to know whether their shared hypothesis was correct. But it has become evident in recent decades, through the emergence of a science sometimes referred to as neurotheology, that the agonies and ecstasies of religious experience are often reflected in the workings and architecture of our brains.

Read more: "Was Jesus a Shaman?"

Recently, an international team of scientists from Europe, Asia, and the Middle East set out to systematically review how different religious practices show up in the brain. Across the millennia, humans have devised many ways to commune with the divine: meditation, chanting, recitation, musical worship, prayer, mediumistic trance. Some of these practices require more devotional attention than others, and some of them can produce altered states of consciousness. The team synthesized the findings of 105 studies spanning 28 years (1998-2026), across eight different religious traditions, and seven different types of imaging.

Their primary finding was that religious and spiritual practices can produce consistent, and in some cases, lasting changes to how the physical brain looks and works, such as altered gray matter or connectivity between regions. Each practice differed in significant ways from the next, but two regions showed up most frequently in the results. The first was the prefrontal cortex, the brain's control center, which handles high-level thinking, decision-making, and emotional control. The second was the anterior cingulate cortex, deep in the middle of the brain, which acts as a bridge between the brain's thinking areas and emotional areas. They published their results in the journal *Brain and Behavior*.

Devotional OM chanting, for example, seems to calm both the brain's emotional alarm center, the amygdala, and other emotion centers

found deep in the limbic system, the team found. But Buddhist chanting activated the amygdala instead, and seemed to tune the dial on other emotional centers in the limbic system. Likewise, how prayer shaped the brain seemed to depend very much on the type of prayer practiced. Improvised personal prayer was found to activate regions of the brain central to interpersonal connection. Another kind of prayer associated with Sahaja yoga, aimed at quieting the mind, forgiveness, and humility, seemed to deactivate certain parts of the brain associated with external awareness and top-down mental control. And people regularly practicing a type of daily Islamic prayer called *Salat*, which includes posture, recitation, breathing, and focus, showed differences in the corpus callosum, which connects the two cerebral hemispheres of the brain.

Mediumistic trance, meanwhile, as measured by electroencephalogram, looked like a distinct and healthy form of consciousness and produced specific patterns in slower brain waves associated with a person entering a deeply absorbed, automatic state like flow. Some results were counter-intuitive. Hindu Vedic priests trained from a young age in memorization and recitation of thousands of Sanskrit stanzas showed significantly greater cortical thickness than matched controls, localized to regions of the brain involved in self-discipline and emotional regulation, as well as pitch, rhythm, and melody. But they did not show changes in the hippocampus, the brain's hub for forming new memories.

While most of the study participants were healthy, a small subset of them suffered from mental health conditions such as depression. The authors suggest that the findings from these studies yield insights helpful in closing the gap in global mental health treatment. Religious and spiritual practices have not yet been reliably shown to treat diagnosed mental health conditions, but they are, they note, "already ubiquitous, freely available, culturally legitimate, and actively engaged in by a substantial proportion of the population." Many of the studies were also small, correlational, or single case studies, so the findings are preliminary—a starting point.

Hildegard von Bingen never had a brain scan, but nuns who came after her have, and their brains, like those of monks, Vedic priests, and other religious practitioners, show differences that seem to track how they spent their lives. If the causal reading holds, it suggests that

humans may have been actively shaping the way their brains work through religious practice since the beginning of time.

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Lead Image: STILLFX / Shutterstock

NAUTILUS

How Menopause Symptoms Could Predict Alzheimer's Risk

JAKE CURRIE · NAUTIL.US · SEP 28, 2026

New research on menopause may explain why women face higher rates of Alzheimer's.

Most people who develop Alzheimer's disease are women. In fact, women account for two-thirds of all Alzheimer's patients—a higher proportion than can be explained merely by their longer lifespans. So what causes the discrepancy?

It's possible that menopause, which men very notably do not experience, could be involved.

During menopause, hormone levels change, with estrogen tapering off as follicle-stimulating hormone rises. These hormonal changes lead to vasomotor symptoms like hot flashes, night sweats, and more. But many women also experience cognitive effects, like difficulty finding the right word or focusing their attention. Additionally, brain regions susceptible to Alzheimer's express receptors for hormones that shift during The Change. Now, a new study published in *Nature Medicine* is uncovering the role this inflection point may play in developing Alzheimer's disease by looking at blood biomarkers.

Read more: "The Evolutionary Mystery of Menopause"

"We don't think that menopause is directly causing dementia," study author Kaitlin Casaletto of the University of California, Santa Barbara explained in a statement. "But it's possible that we may be able to predict a woman's risk for dementia decades later by the levels of these molecules around menopause."

Casaletto and a team of fellow neuroscientists recruited 80 women in their 40s and 50s and determined where they were in their menopausal transition with a measure that combines symptoms and hormone levels. They then tested the women's blood for hundreds of biomarkers and identified 16 whose levels were associated with menopausal changes. Importantly, these molecules were more closely linked to shifting hormone levels than the women's ages, a strong suggestion the changes were related to menopause.

As estrogen levels fell, the researchers discovered proteins involved in inflammation pathways rose. This coincides with a general increase in postmenopausal inflammation other researchers have found that could set the stage for later neurodegenerative diseases. As follicle-stimulating hormone levels surged, levels of other proteins involved in synaptic function also climbed. These proteins are biomarkers for Alzheimer's disease as well.

Of course, 80 women isn't the largest sample size, so the team analyzed health data from more than 2,800 women enrolled in the United Kingdom's BioBank study to confirm their findings. They then widened the scope of their investigation, looking at data from over 12,000 women in their 60s and 70s involved in four other studies. They found that women with the highest levels of these proteins had both poorer memories and a 15 percent higher risk of developing Alzheimer's disease later on.

The team uncovered links between vasomotor symptoms and these biomarkers as well. Women who reported night sweats during menopause also showed a bigger increase in inflammatory proteins. Additionally, older women who remembered experiencing hot flashes had higher levels of these proteins much later in life, which suggests a lingering effect.

It's not all bad news, though. According to the team, these findings could pave the way for future blood tests that can act as an early warning system for women who are most at-risk. "Midlife is a critical window for both men and women where we can really set up the brain to be more resilient or more vulnerable to dementia in later life," said Casaletto. "For women, menopause may be this especially powerful time to intervene."

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Lead Image: Ruslana Naida / Shutterstock

NAUTILUS

Famed Dancer's Diary Offers Glimpse into Psychosis

JAKE CURRIE · NAUTIL.US · SEP 28, 2026

A rare first-person record of a descent into psychosis, preserved in a dancer's own diary.

Called the "god of dance," Vaslav Nijinsky was one of the most widely renowned dancers of the 20th century. He gained fame as a male ballet dancer and choreographer in the celebrated Parisian dance company, Ballets Russes. Unfortunately, his brilliant career was cut short by a diagnosis of schizophrenia. He was institutionalized in 1919 and spent the last three decades of his life in and out of asylums. During a six-week period when he was gripped by psychosis, he kept a diary, which is now the subject of a new study into how mental illness can affect language.

"Language provides a window into someone's mind," explained study author Alban Voppel, professor of AI and psychiatry at McGill University in a statement. "Complete book-length texts written during an active psychotic episode are quite rare, giving us a unique opportunity to study language as the illness was unfolding."

Read more: "Fire in the Brain: How Psychosis May Arise When the Body Attacks Itself"

Voppel and a team of researchers conducted a computational linguistic autopsy of Nijinsky's diary using tools that measured patterns in word use as well as sentence structure. They wanted to find out if the linguistic pathologies of formal thought disorders (like schizophrenia) manifest in how topics are connected or in the words used to convey them.

To find out, they compared Nijinsky's writing to a control text from his wife Romola: the forward she wrote when she published a heavily edited version of his diary in 1936. They chose her writing because the two texts were of similar length, from roughly the same period, and written about similar subjects. Most importantly, Romola had no history of psychiatric illness.

The team found Nijinsky often wrote in circles using the same words repetitively. This had the effect of demonstrating a kind of superficial coherence, when the work as a whole was generally incoherent. "It's sort of a fake coherence, a mechanical coherence," Voppel said. "He

came back to the same topics and used a lot of the same words, especially 'I,' 'God' and 'love.'"

These words represented a kind of structural scaffold for the rest of Nijinsky's meandering thoughts—the lettuce in his word salad.

According to the team, this suggests that psychosis can affect language at different levels. It's possible that a person succumbing to acute mental illness can still communicate in a broad narrative structure while their vocabularies become increasingly limited by their disorder. "Nijinsky's diary is not simply a document of suffering or a record of thought disorder," the researchers wrote, "it is an unwitting self-portrait of a mind condensing around a fixed, narrow repertoire of words and the concepts they stand for."

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Lead Image: Karl Struss / J. Willis Sayre Collection of Theatrical Photographs / Wikimedia Commons

NAUTILUS

How Snail Shells Could Be Used to Map Prehistoric Cyclones

DAVID NIELD · NAUTILUS · SEP 29, 2026

Growth rings in ancient snail shells may hold a record of storms from thousands of years ago.

The hard shells around snails protect them against predators, act as cover for rain and sunshine, and lock in moisture. They're also able to serve as meteorological time capsules, a new study shows.

Researchers examined a shell from a snail that died in August 2018, in the Coalstoun Lakes National Park in South East Queensland, and identified it as a Biggenden Banded Snail (*Figuladra bayensis*). Then, using radiocarbon dating and oxygen and carbon isotope analysis, they determined that the snail had lived for four-and-a-half years. Its shell also showed signs of rapid growth spurts, together with periods of slow growth, or no growth at all.

Read more: "How Snails Cross Vast Oceans"

The normal expectation would be that these spurts correlated to wetter seasons, when there was more food around for the snail to languidly snack on, but the researchers discovered that the real story was more dramatic.

"The results showed the snail was responding to extreme rainfall events rather than annual wetness," study author Jamie Shulmeister said in a statement. "Because we knew when the snail lived, the growth spurts could be linked to periods immediately following cyclones in 2015 and 2017."

These cyclones are well-documented: Severe Tropical Cyclone Marcia and Tropical Cyclone Debbie respectively. As Marcia and

Debbie left the forest floors saturated, snails like the one studied were able to drink deep, imbibing the ultralight oxygen that characterizes the water in fast-moving storms.

Shells, it turns out, work a little like tree rings: They lock in some of the environmental conditions of the time, so scientists can come back later and read the signs etched into the calcium carbonate.

This is just a single shell, in a single location, but if the same analysis can be applied to others, we've got a new way of identifying the times and places of storms back through the centuries. In fact, shells could take us much further back in history than conventional meteorological records.

Matching older samples would be harder and less precise, but they could answer questions about the frequency or intensity of cyclones and establish the climate records of areas over time.

"It was surprising and quite exciting that the humble snail could be a tool to reconstruct the paths of past cyclones even in prehistoric times," study author Nicholas Patton said. "If we're able to find shells within different layers of sediment deposits, we can combine the individual shell records together to extend this information further back in time."

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Lead Image: Agita Leimane / Adobe Stock

NAUTILUS

Frogs Wear Bacterial Armor to Defend Against Fungus

DEVIN REESE · NAUTILUS · SEP 28, 2026

A protective microbe layer is helping some frog populations survive a fungus that is wiping out others.

Unlike scaly reptiles, frogs aren't well-armored. But a recent study published in the journal *Animal Microbiome* shows that some frogs do sport hidden armor for defense in their battle against pathogens.

One of frogs' fiercest enemies is a deadly fungus that colonizes their skin and eventually causes heart failure. This chytrid fungus (*Batrachochytrium dendrobatidis* or Bd) has devastated amphibian populations worldwide, driving dozens of species to extinction.

Read more: "The Secret Lives of Japanese Temple Frogs"

At the same time, many frog species somehow tolerate it. Case in point: In the rainforests of Brazil, Hensel's big-headed frog (*Ischnocnema henselii*) is highly susceptible to the fungus, while the ecologically similar clay robber frog (*Haddadus binotatus*) is not.

"For years we knew that the tropical frog *H. binotatus* rarely became infected with the fungus in the wild, but we didn't know why," said lead author Laura Schuck, a doctoral student at Penn State.

Schuck and her colleagues in Brazil collected 40 individuals of each species from forest fragments near São Paulo. Back in the lab, half the frogs in each species were exposed to the Bd fungus while the

other half were not. In addition, half the frogs in each group received antibiotic baths that killed their bacterial skin microbiomes, or "bacteriomes," before fungal exposure, while the other half did not.

Clay robber frogs survived well when exposed to the Bd fungus but, if bathed with antibiotics first, their survival plummeted. Hensel's big-headed frogs fared poorly, regardless of antibiotic treatment. So, the unique bacteriome of clay robber frogs, but not of Hensel's big-headed frogs, acts as protection against the Bd fungus. "This study moved us from correlation to experimental evidence that the microbiome really is important against this disease," remarked Schuck.

Schuck also extracted about 700 bacterial strains from the frogs' skin, then grew each one in culture and exposed it to the Bd fungus. After scoring how each bacterial strain affected fungal growth, she sequenced the cultures for DNA and mapped them relative to their effects on the fungus to create a reference database.

The database will be like an armory—the ultimate guide to which frog species have the necessary defense to withstand the Bd fungus.

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Lead Image: Renato Martins / Penn State

NAUTILUS

A Dispatch from the Trenches of Algorithmic Living

NICOLE A PRUCHA · NAUTIL.US · SEP 28, 2026

An essay on what it feels like to have algorithms quietly curate your everyday choices.

This excerpt is reprinted with permission from NYU Press. It is adapted from *Power in Listening: The Sounding Out!* Reader.

Sometime in 2021, I opened the Spotify app, navigating my way to a recent playlist I had made featuring Arlo Parks, Brittany Howard, Hozier, Charlotte Day Wilson, Radiohead, and other artists known, at least in part, for their melancholic sounds and senses. Immediately, under the banner "You Might Like," was a featured playlist titled "sad girl starter pack."

As is common in the age of targeted ads, privacy agreements, and social media, I felt simultaneously exposed and validated. As someone who surveys the personalized "Daily Mixes" and "Discover Weekly" playlists, searching for new music daily, it seemed like the perfect addition to my growing playlist collection. Here were 75 songs, some by artists I knew and others I didn't, that at first glance seemed tailored to my taste. But, the longer I stared at my screen, my excitement turned into skepticism.

Read more: "When You Listen to Music, You're Never Alone"

Something was off. It, like many of Spotify's mood-based playlists, claimed to be curated based on a sense, as opposed to a genre, decade, or other category marker. However, unlike the application's popular mood-based playlists (e.g., "Feelin' Good," "All the Feels," "Broken," "Chill Vibes"), this playlist evoked both gender and affect, in its name alone. Clearly, it was meant to conjure, capture, and attract a specific version of both femininity and sadness through sound. But what was that version? And what did it, supposedly, sound like?

Since first pressing play on Charlotte Day Wilson's "Work," one of the featured tracks, Spotify's "sad girl starter pack" has continued to haunt me. Early into 2023, Spotify launched a global music program titled GLOW, celebrating and promoting LGBTQIA+ artists and creators. This new digital music hub boasts "Sounds from the Community"—playlists featuring artists who identify as LGBTQIA+—as well as a collection of playlists deemed "LGBTQIA+ Favorites." Among this latter collection is none other than "sad girl starter pack," which also happens to be the hub's most-liked playlist, by far, with over 1.3 million likes. Seeing it there, I once again felt attacked, but from a different angle. To a queer woman who still loves listening to slow, sad songs, how could the same playlist that once raised questions about sadness, femininity, and sound manage to provoke new questions about queerness, too?

In addition to being in the GLOW hub, the playlist's new description also unmistakably invokes queerness. Two years ago, when the playlist was first released, the description read, "Is her favorite color sage green?" A nod at one of the trendy colors that defined late 2021 and 2022, the description feels playful, harmless, and more so a reference to an aesthetic than a gesture toward the rainbow of identities constituting queerness. Later on in 2022, the description remained somewhat obscure, but became more pointed: "Sapphic songs that defined your music taste as 'yearning.'" The word "sapphic," derived from the poet Sappho and often used to describe sexual and romantic attraction between women, clearly suggests queerness. And yet the playlist is not explicitly queer, at least not in an identitarian sense. That is, it does not feature exclusively queer artists like the playlists in GLOW's "Sounds from the Community"

section. It is also not exclusively feminine in that the included artists are not all women-identifying. In fact, as a personalized playlist—Spotify’s hybrid between an editorial and an algorithmic playlist—some of its songs change from user to user.

What seems to be the unifying theme of the playlist, then, is a soft, melancholic sense of yearning that is descriptive of both the music in the playlist and the audience clearly targeted by it. According to the packaging, marketing, and evolution of “sad girl starter pack,” that audience appears to be sad girls, women, and femmes, queer or not. But this collapsing of identities into a single sense, category, or even playlist can be a dangerous game.

Spotify: The Sad Girl’s Digital Domain

As a digital platform, Spotify clearly understands and capitalizes on the sociality and circulatory nature of music. The “Friend Activity” column on the desktop application allows speculation about our friends’ moods based on what they’re listening to; Spotify’s annually anticipated marketing campaign “Wrapped” is designed for sharing across social media platforms with ease; and the app’s “Blend”

feature can create a personalized playlist combining up to ten friends’ music tastes. These features underscore something we—music lovers, listeners, and consumers—have always known: Listening is done in, for, and about community. Playlists, arguably Spotify’s defining feature, are thus powerful social tools of relation, no matter their curator. To some, they are epistolary, sent between crushes like vaguely opaque love letters. To others, they are time capsules, relatives of the cassette tape and CD mixtapes that conjure memories and stories. For still others, they are the sonic backdrops of study sessions, breakups, warm-ups, parties, and more. Regardless of how they’re perceived, playlists are inherently relational and deeply affective. It is no wonder then, that “sad girl starter pack” is as popular as it is. Not only does it implicate multiple communities in its title and description, it invites listeners from those communities and beyond to listen, feel, and—to use the language of the playlist—yearn together. But, as this chapter’s opening foreshadows, this rosy depiction of the playlist’s social potential, through affect, begins to unravel as we consider who is hailed by this playlist and how.



Continued online (full article about 2646 words). Scan, or visit <https://nautil.us/a-dispatch-from-the-trenches-of-algorithmic-living-1285345/>

LONGREADS

Dusklands

SEYWARD DARBY · LONGREADS.COM · SEP 29, 2026

An essay on growing up in the long shadow of genocide.

Celebrated novelists J.M. Coetzee and F.M. Siccardi recently published a co-written book, *Un mal salvaje* (A Savage Evil), about their experiences coming of age in South Africa and Argentina, respectively. Both states were created by colonial forces that all but wiped out indigenous populations and cultures. This excerpt juxtaposes the authors’ childhood memories of life in the shadow of genocide, though neither would have known to call it that when they were young. In particular they recall the workers on their families’ land—men they somehow knew both intimately and not at all:

Freek did not see himself as a tragic figure, did not even see himself as a survivor. To see yourself as a survivor of the past you must have a firm conception of that past, you must know where you come from; and Ros and Freek had no such historical consciousness. First, they

were illiterate. Second, it was of the essence of the genocide which they had survived that communal memory was erased.

If I were magically able to return to the 1950s and tell Freek about the Khoi who had pastured their stock at the Cape for hundreds or perhaps thousands of years before the Dutch came and laid waste to their society, about a people who were his ancestors, he would have thought it was just a story I was making up. Genocide is not just the destruction of a people, it is also the obliteration of that people’s store of memories and, with it, the destruction of their ability to see themselves historically.

More stories about South Africa

“A yearlong investigation into the African-grey trade reveals a web of poachers, egg smugglers, wealthy businessmen — and multitudes who want a talking bird.”

PSYCHE

When you feel moral disgust, what's the emotion telling you?

BRANDON YIP · PSYCHE.CO · SEP 29, 2026

A look at whether gut-level moral revulsion is a reliable guide or just a reflex.

‘Disgusting’ and similar words are flung across political divides quite often today. For example, the US president Donald Trump has used the word to describe his political opponents, journalists and others. And many have voiced disgust at the president’s own behaviours. For instance, after viewing Trump’s

altercation with the Ukrainian president Volodymyr Zelenskyy during a press conference in 2025, the author David Brooks described feeling ‘nauseated, just nauseated’. In politics and beyond, expressing disgust is a common response to actions that one finds

deeply objectionable, such as racist rhetoric, acts of sexual abuse or policies that harm vulnerable people.

Rather than see expressions like these as merely rhetorical, many philosophers and psychologists argue that moral disgust is a distinct and significant kind of moral response. The distinction between moral disgust and regular disgust lies in the way their objects are experienced by the disgusted person. People feel regular disgust in response to the spread of filth, which motivates physical avoidance and cleansing. The stench of urine on a train, for example, provokes this kind of disgust. When people feel moral disgust, it's as if they are experiencing a sort of moral pollution, which motivates them to socially avoid and exclude the target of the disgust. Betrayal and exploitation, for example, often provoke moral disgust, but not regular disgust.

Many scholars believe that disgust operates based on simplistic and unreliable heuristics. For example, in his book *Yuck!: The Nature and Moral Significance of Disgust* (2011), the philosopher Daniel Kelly argues that disgust is an evolutionarily old mechanism that protected humans from poisons and parasites – and so it operates on a 'hair trigger' that is prone to generate false positives. Kelly thinks that, since moral disgust grew out of regular disgust, it is similarly unreliable when it comes to moral matters. Just as you can find some physical thing disgusting (say, a certain kind of cheese) when it's actually harmless, you might automatically find some behaviours morally disgusting that, on reflection, aren't so bad.

Now, some of Kelly's premises can be challenged: there is evidence that our emotional systems are intelligent and adaptable. Rather than just recapitulating ancient functions, our emotions contain updating mechanisms that enable them to respond to new physical and social challenges. Furthermore, even regular disgust is flexible and remarkably sensitive to context. For example, whether you find saliva disgusting depends a lot on whether it comes from your lover or a total stranger.

Some have argued that, far from being unreliable, moral disgust plays an invaluable role in moral judgment. In Leon Kass's classic essay 'The Wisdom of Repugnance' (1997), he claims that people's disgust towards human cloning gives us some reason to avoid that practice, warning us 'not to transgress what is unspeakably profound'. In this decadent age, Kass says, 'repugnance may be the only voice left that speaks up to defend the central core of our humanity.'

So, is feeling morally disgusted by an act or a practice something that we should take seriously as a sign that it is morally problematic? To answer this, we need to first get clear on why people feel moral disgust – what exactly it alerts us to.

Contemporary philosophers have offered various explanations of what moral disgust does, though each of these has limitations. In Alexandra Plakias's account, for example, moral disgust is a response to behaviours that are socially contagious: norm-violating behaviours, such as acts of physical violence, that might spread through a population if left unchecked. This is a compelling idea, but upon further examination, the relationship between socially contagious behaviours and moral disgust appears to be tenuous. Many norm violations that could be socially contagious – such as stealing office supplies or not declaring tips as taxable income – don't tend to strike people as morally disgusting.

Presenting a different view of moral disgust, Victor Kumar suggests that it's a response to behaviours that undermine relations of trust and reciprocity. He is certainly right that behaviours that undermine cooperative relations, such as betraying one's compatriots or unfairly dividing rewards, are often seen as morally disgusting. Yet there are also many other acts that people find morally disgusting – such as cannibalism, or incest among consenting adults – that do not involve a failure of reciprocity.

Imagine you're in a soccer game with players who are randomly picking up the ball and throwing it around

I believe that John Kekes was on the right path with a suggestion he put forth in the 1990s. Kekes proposed that moral disgust enforces moral taboos – prohibitions against the kinds of actions that endanger the very possibility of civilised life. These violations supposedly 'threaten to bring about a form of life in which moral considerations have no foothold.' This seems somewhat overblown: after all, civilisations have repeatedly survived the violation of moral taboos. But perhaps what moral disgust does is guard against the collapse of particular forms of rule-structured life.

To see what I mean, consider the relationship between our personal behaviours and the social worlds that we inhabit. A social world is structured by rules and norms that people have internalised and adopted. Tamar Schapiro points out that these normative orders are sustained by individuals' compliance with the rules. A judge who continues to follow legal procedure – even when they could choose to act corruptly – helps to sustain the larger normative structure that gives the judge's actions legitimacy.

If norm compliance keeps important social structures intact, then perhaps certain actions can undermine them. Imagine that you are trying to play soccer with players who are randomly picking up the ball and throwing it around. Such behaviour not only violates the rules of the game, it undermines the game itself. Something analogous happens in typical cases of moral disgust, where the stakes are higher. In explaining his nausea at seeing Trump behave 'vilely' toward Zelenskyy and US allies, for example, Brooks suggested that Trump was unravelling a benign, US-centred international order and replacing it with one where 'might makes right'. Let's call actions that undermine a social order socially corrosive.

My proposal, then, is this: moral disgust is a response to behaviour that is socially corrosive. More precisely, when some behaviour undermines a set of moral norms that a person is attuned to, they feel morally disgusted.

The moral norms you are attuned to are those that appear to you to impose genuine constraints on behaviour. Your attunement to these norms might be strong and overt – you might explicitly endorse your culture's norms against using corporal punishment on children, for example. Or, your attunement might be merely implicit. If you grew up in an Islamic culture that sees eating pork as morally disgusting, but then you left Islam, you might still continue to experience the pork prohibition as having some force. People can also become attuned to new norms. An adult Muslim convert – or a vegan – could come to acquire moral disgust towards pork-eating.

Behaviours once seen as morally disgusting, such as premarital sex, are now widely seen as acceptable

When I suggest that morally disgusting actions undermine some social order, I mean that they threaten to stop a community from operating with some set of moral norms. For example, if a political leader engages in corruption or an employer openly flirts with his employees, then the norms that govern that society or that workplace will be undermined as long as such behaviour goes unchecked. This possibility is more serious than just contagious norm-violation; it is not simply that more corruption or more creepy flirting is forthcoming. It's that the rule of law or respectful workplace interactions become tenuous altogether. Moral disgust alerts us to the fact that certain behaviours might corrode the moral norms that we care about, and it motivates us to sanction or exclude those who behave in those ways.

With this in mind, it is worth returning to Kass's article about cloning and disgust. He notes that human beings have 'systems of identity and relationship on the bases of ... deep natural facts of begetting,' and suggests that cloning would confound 'all normal understandings of father, mother, sibling, grandparent, etc.' To Kass, then, what is problematic is not just the wrongness of cloning in itself or its mere proliferation. Rather, accepting cloning would unravel a set of norms governing human relations.

Could cloning really undermine all that? It's not as implausible as it might seem. Already, the norms surrounding sex and the family have undergone radical transformation over the past century. Behaviours once seen by many as morally disgusting, such as premarital sex or homosexuality, are now widely seen as morally acceptable. This has produced profound changes in the meaning of marriage and sex. Even if we consider these changes liberating – welcome alterations to an overly constricting social order – these examples show how it's plausible that a broader set of moral norms (such as those structuring families) is susceptible to change once key norms (such as those governing procreation or sex) become undone.

Clearly, we should not uncritically endorse the deliverances of moral disgust. One may, for example, decide that the familial structure that Kass protects is an unjust and patriarchal one. The norms we are attuned to are a mixed bag that includes both aspirational ideals that bend towards justice, as well as the nastier parts of our cultures.

Yet it is still worth listening to our moral guts – carefully. Moral disgust sometimes provides a signal that something is deeply wrong, even when one's capacity for moral reasoning has been compromised. Ditte Marie Munch-Juriscic has noted, for example, that the perpetrators of heinous crimes often react with disgust toward their own actions. Under the Nazi regime, some soldiers and citizens experienced sheer revulsion in carrying out their role in the Holocaust. They suppressed this revulsion with reasons provided by the ideological narratives of the Nazi regime. Similarly, Miranda Fricker has shown that injustice sometimes makes it hard to understand the conditions of oppression. For example, prior to acquiring the concept of sexual harassment, many women struggled to articulate the problematic nature of men's inappropriate advances. But they were morally disgusted by such behaviour, even before they could fully explain why it was wrong – including how it undermined norms of equal respect for men and women.

If there is a wisdom in moral disgust, then, it is an ambivalent one that we should not always heed, yet cannot simply ignore. Unfortunately, there isn't a simple formula to tell us whether moral disgust is highlighting something important or leading us astray. When we feel or encounter moral revulsion, we should neither dismiss it as mere primitive reaction nor treat the emotion as having an infallible moral authority. Instead, we ought to pause and reflect on what we are actually responding to. We should try to articulate the normative order that our revulsion is guarding – and then ask ourselves whether or not we have good reasons to cherish it.

LONGREADS

Lost and Found: A Reading List on Labyrinths, Mazes, and Wayfinding

KANYA KANCHANA · LONGREADS.COM · SEP 29, 2026

A meditative reading list on the difference between being lost and being found.

Thoughtful stories for thoughtless times. Longreads has published hundreds of original stories—personal essays, reported features, reading lists, and more—and more than 14,000 editor's picks. And they're all funded by readers like you. Become a member today.

Once, as a young adult in the city, I was walking home after a long day of work. The main street leading to my place wasn't well lit, with dark stretches between streetlamps. There was a huddle of men under one, some standing around, some lounging against the post. They

moved in the orange light, turning as one to face me. Something shifted in the air.

An earlier version of me would have been slower to respond, given them the benefit of the doubt. But after a recent knife-point mugging, I had begun to develop a sort of second skin—a keen, physical sense of danger. I slipped back into the shadows and ran in and out of alleyways. I doubled back, cut across fences, and scaled half walls. I made no sound at all; I made it home. I had made a spiraling maze in real time. They could not find me.



Continued online (full article about 3631 words). Scan, or visit <https://longreads.com/2026/09/29/labyrinths-mazes-kanya-kanchana/>